

# Net-Zero Equity Strategy — European Region

A decarbonisation mandate for institutional investors

## Client Objective

Help institutional investors reduce the carbon footprint of their European equity portfolios while preserving a diversified, benchmark-aware allocation. The strategy supports pension funds, insurers, and wealth managers in aligning portfolios with climate-transition objectives without sacrificing risk-adjusted performance. It targets ownership-based emissions (Scope 1 + Scope 2) and tracks the value-weighted European equity benchmark closely.

## Why It Matters

Carbon-intensive firms face rising transition risks — regulation, carbon pricing, and shifting investor preferences. Our analysis shows these risks can be addressed without abandoning financial targets. The net-zero glide path reduces the 2024 carbon footprint to **58.7 tCO<sub>2</sub>e/M\$ invested**, a **71.8% reduction** from the 2013 EUR benchmark base-

line of 207.92 — delivered at a realized tracking error of only **0.38%** and with a Sharpe ratio fractionally above the benchmark itself. Decarbonisation becomes measurable, credible, and financially manageable.

## Strategy Overview

A three-step process: (i) measure the carbon footprint of the European equity benchmark, (ii) reduce exposure to the largest ownership-weighted carbon contributors via tracking-error minimisation, (iii) reallocate capital toward financially comparable, lower-emission firms. The net-zero variant tightens the carbon constraint by 10% per year from the 2013 VW baseline, defining an annual glide path:

$$CF_Y^{\text{target}} = (0.90)^{Y-2012} \times 207.92.$$

By 2024 the target reaches 58.7 tCO<sub>2</sub>e/M\$ invested.

## Return preserved ■ Risk controlled ■ Carbon reduced

**8.18%**

Annualised return  
of the net-zero strategy  
(VW benchmark: 8.19%)

**0.38%**

Realized tracking error  
vs. VW benchmark  
(IR near zero)

**0.407**

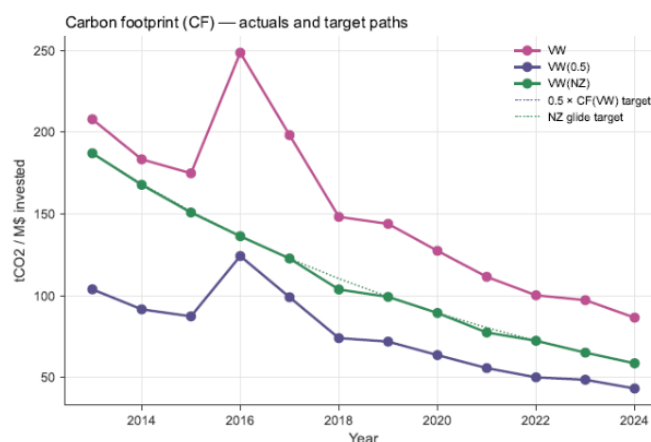
Sharpe ratio  
(VW benchmark: 0.406)

**71.8%**

Carbon footprint cut  
vs. 2013 VW baseline,  
by 2024



Net-zero remains close to VW in cumulative performance.



Net-zero follows a disciplined declining carbon-footprint path.

## Trade-offs & Risks

### Tracking-error risk & sector tilts

Decarbonisation requires underweighting carbon-intensive sectors (energy, utilities, materials). The benchmark-aware framework caps the resulting deviation: realized tracking error stayed at 0.38% annualised over 2014–2025. Active sector exposures are monitored quarterly.

### Constraint feasibility

The net-zero glide path was feasible in all 12 rebalancing years over 2013–2024. In four of those years (2014, 2015, 2018, 2021) the EUR benchmark itself already met or beat the target — the constraint added no friction. As the path tightens beyond 2025, the supply of clean substitutes narrows and warrants ongoing review.

### Data quality & Scope 3 exposure

The analysis uses Scope 1 + Scope 2 emissions only. Scope 3 can be material in financials, technology, and retail. Carbon and revenue data are forward-filled when missing, creating stale-observation risk. Mitigation: quarterly data refreshes and explicit flagging of firms with high reporting uncertainty.

### Carbon coverage risk

A lower portfolio carbon footprint does not capture every dimension of climate exposure (physical risk, biodiversity, transition-plan credibility). The strategy is designed to be combined with ongoing monitoring of issuer-level transition plans and improvements in disclosure quality.